LAYER DECOMPOSITION

DLCV Fall 2025 Final Project - Challenge 2 (PicCollage)

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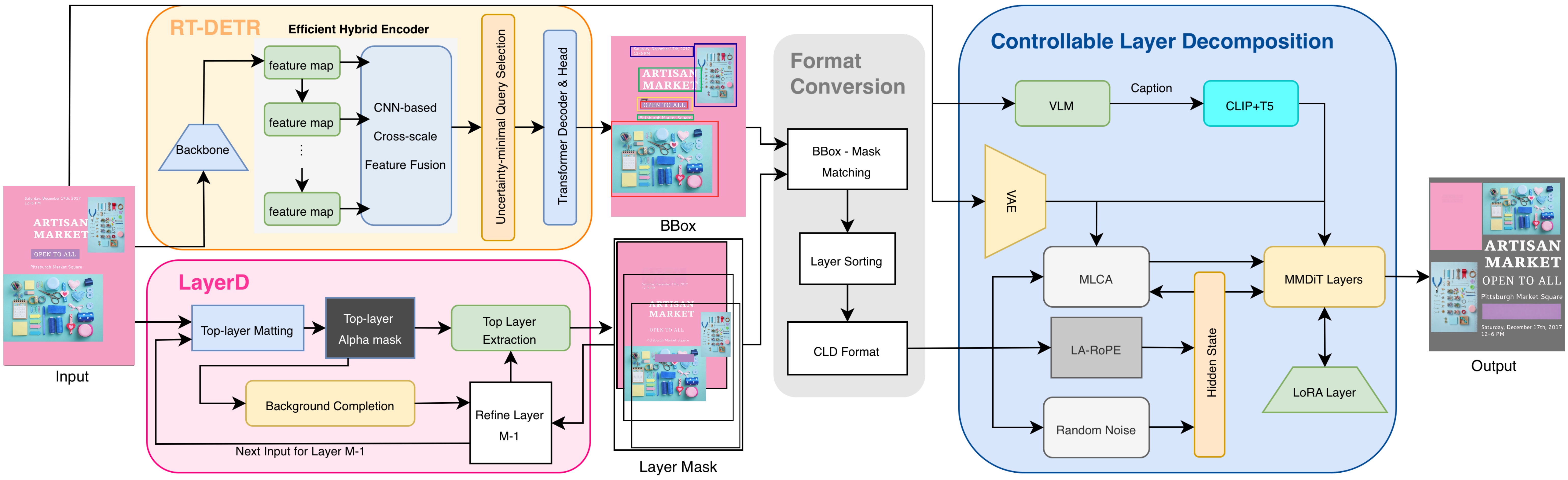


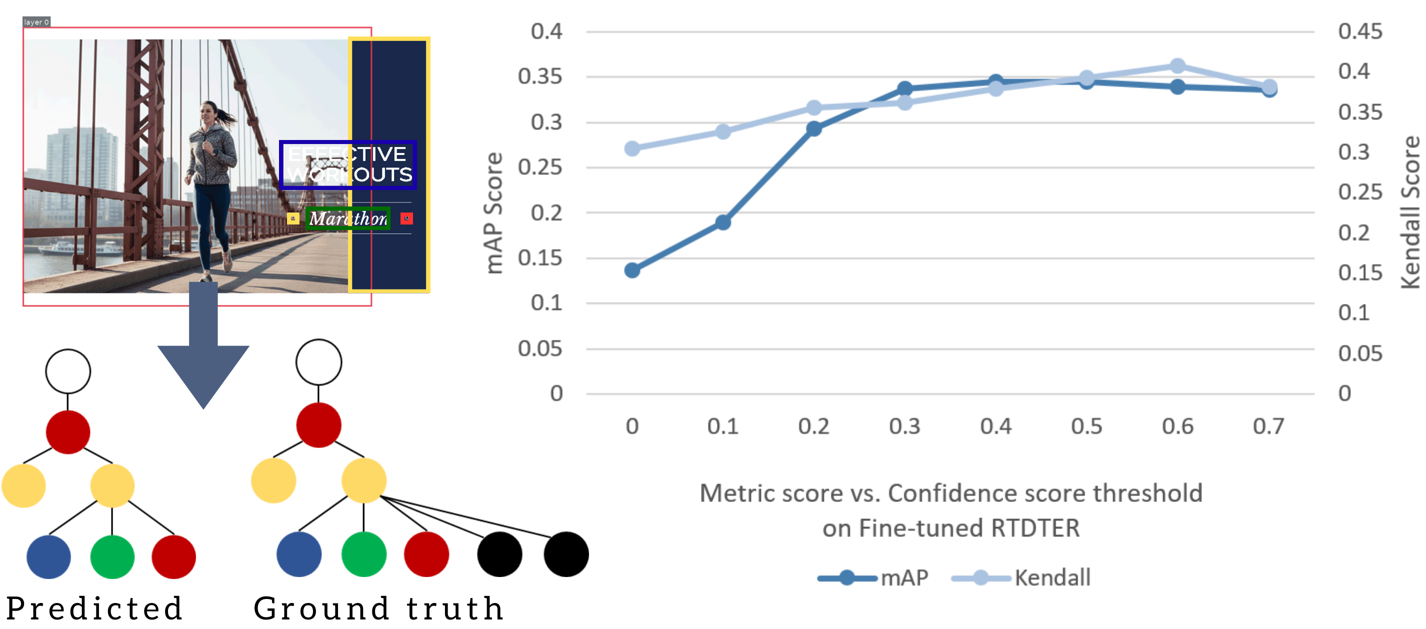
METHODOLOGY

Our modular pipeline comprises four key stages to achieve layer decomposition on input flattened image. (1) First stage perform object-level localization using RT-DETR, which detects and outputs bounding boxes for elements. (2) Second, LayerD executes iterative decomposition to generate masks for each element and the background. (3) Next, a conversion stage aligns bounding boxes with masks while optionally using LLaVA to generate descriptions. (4) At last stage, CLD model synthesizes the final RGBA layers, ensuring each layer is visually complete and order-consistent for final rendering.

EVALUATION

We use Kendall distance via Hungarian matching and mAP@0.5 as our metrics on the bounding box and layer info predictions. We further introduce a normalized Tree Edit Distance (TED) that addresses the problematic situation that limits the previous metrics. Furthermore, a reference-free IQS is introduced to quantify layout sanity by integrating confidence.

We will use these metrics to evaluate our various experiments, including different confidence thresholds and detector models trained with diverse inputs such as depth maps and captions.



(a) Trees for the image

(b) Finding the most suitable thresholds

Fig 1. (a) The trees are established through bounding box containment and spatial overlap. Their similarity is calculated for evaluation. (b) Evaluate our detector with different confidence score thresholds

EXPERIMENTS & RESULTS

	mAP	Kendall	TED (unit)	TED (area)	IOU
RT-DETR	0.3448	0.3790	0.0986	0.6623	78.01
RT-DETR with depth	0.3515	0.4204	0.1415	0.6812	78.67
RT-DETR + LayerD + text	0.3087	0.2348	-0.0054	0.6660	79.70
RT-DETR + LayerD	0.3427	0.3647	0.0972	0.6622	79.20

Table 1. The results of various experiments.

Here we demonstrate the outputs generated at each stage of the pipeline:

Fig 2.(a) Original input image

Fig 2.(b) RT-DETR BBox

Fig 2.(c) LayerD Mask

Fig 2.(d) CLD reconstruction

Our pipeline achieved final score of 79.63, calculated as weighted combination of 0.1*global reconstruction loss + 0.9*mean per-layer loss.

CONCLUSION

We developed a pipeline integrating RT-DETR localization with LayerD and CLD generative power. By modeling spatial structure and compositional consistency, our system effectively decomposes images, reconstruct elements, and inpaint hidden layers. PicCollage benchmark results demonstrate effectiveness